Modelling the Annual Frequency of Extreme Temperature Events in the United Kingdom, 1901–2023

MATH11188 Statistical Research Skills — Group 15

1. Introduction and Research Question

Extreme temperature events have significant consequences mainly in public health, infrastructure, and the ecosystems [3]. Since the Scottish Government has increasingly focused on climate adaptation, quantifying temporal trends in extreme event frequency is of direct policy relevance. Hence, the question has been addressed: **Which count regression model Poisson, Quasi-Poisson, or Negative Binomial best describes the annual frequency of extreme temperature months in the UK (1901–2023), and is there statistical evidence of a temporal trend?**

2. Data and Pre-processing

Monthly temperature data were taken from the gridded dataset CRU TS v4.08 [1], which covers the period 1901–2023 with a spatial resolution of 0.5°. From the total grid, only points within the geographical boundaries of the United Kingdom were selected (longitude −8.5 to 2.0 and latitude 49.5 to 61.0 21×23 points) from which we calculated the average temperature for the whole UK for each month (Figure 1).

To define what is expressed as an extreme event, we followed the IPCC AR6 WG1, Chapter 11 [3], which states that a month can be characterised as extremely warm if its temperature exceeds the 95th percentile, or as extremely cold if it is below the 5th percentile of the typical climatic period given by the World Meteorological Organization(WMO) 1961–1990. Applying this theory specifically for the UK, we resulted in fixed thresholds where for an extreme warm month, the temperature should exceed 14.72°C and for an extreme cold month it should be below 2.71°C.

These thresholds were applied uniformly over the entire time period 1901–2023 so that we can see if extreme warm/cold months are increasing due to climate change. The period 1961–1990 is used as a baseline first to avoid always having the same percentage of extreme months, secondly to be able see potential trends.

Finally, a binary indicator was defined for each month (1 if the month was extreme and 0 otherwise) which were summed at the annual level to obtain the number of extreme months per year Y_t (values $0, 1, 2, \dots$) count data modeled with Poisson-type models, while the variance-to-mean ratio of $0.922 \approx 1$ indicates approximately Poisson uniform dispersion.

3. Statistical Models

By using Y_t as the annual count of extreme months in year t, with exposure $m_t = 12$ months, we fit three GLMs with log link:

$$\log\big(\mathbb{E}[Y_t]\big) = \beta_0 + \beta_1\, t + \underbrace{\log(m_t)}_{\text{offset}}, \qquad t = 1901, \dots, 2023, \tag{1}$$

The **Poisson**,**Quasi-Poisson** and the **Negative Binomial** all of them have as main difference their variance structure. The **Poisson** model assumes $\mathrm{Var}(Y_t) = \mu_t$ (equidispersion). The **Quasi-Poisson** model estimates a dispersion parameter $\hat{\phi}$, yielding to $\mathrm{Var}(Y_t) = \hat{\phi}\,\mu_t$, that corrects the standard errors of the coefficients, but without changing the estimates of the coefficients themselves (β_0, β_1). The **Negative Binomial** introduces a new parameter $\theta > 0$ with $Var(Y_t) = \mu_t + \mu_t^2/\theta$, which allows the variance to exceed the mean (overdispersion). If $(\theta \to \infty)$, then the second term approaches zero and the model effectively reduces to Poisson model.

4. Model Selection and Goodness-of-Fit

4.1 Relative Goodness of Fit

The model comparison criteria are summarise in Table 1. Among all models,the Poisson model achieves the lowest AIC (344.18) and BIC (349.81), indicating superior parsimony. Although,the Negative Binomial has higher AIC (346.18) and BIC (354.62), and the likelihood ratio test gives $p = 1.0$, indicating that there is no evidence that the additional dispersion parameter improves the fit. The estimated Quasi-Poisson dispersion with $\hat{\phi} = 0.903 \approx 1$ confirms the existance of equidispersion, and the Cameron–Trivedi overdispersion test yields $p = 0.850$. The Negative Binomial theta estimate ($\hat{\theta} = 17{,}956$) is extremely large,as a result is confirming the convergence to Poisson.

4.2 Absolute Goodness of Fit

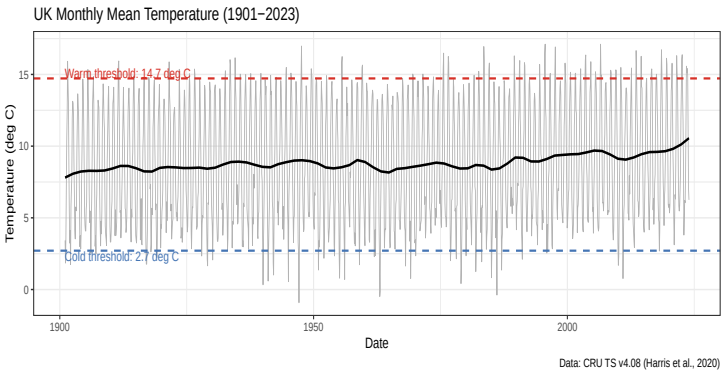
All three models with 121 degrees of freedom have the same Pearson chi-square tests $\chi^2 = 109.20$ and (p=0.771), indicating that the observed counts are statistically consistent with the model predictions.This is expected as all three models share the same mean structure, and thus the fitted values generated were similar. Additionally, DHARMa simulation-based residual diagnostics [2] (Figure:d) confirm adequate fit for the Poisson model (KS uniformity: $p = 0.596$, dispersion: $p = 0.300$) and Negative Binomial (KS: $p = 0.667$, dispersion: $p = 0.296$), proving that no significant departures were found. The **Poisson GLM is selected** as the most parsimonious adequate model that fits the data better according to Information Criteria.

Table 1: Relative and absolute goodness of fit measures for the three regression models. DHARMa diagnostics are unavailable for Quasi-Poisson because simulation is not implemented for quasi families.

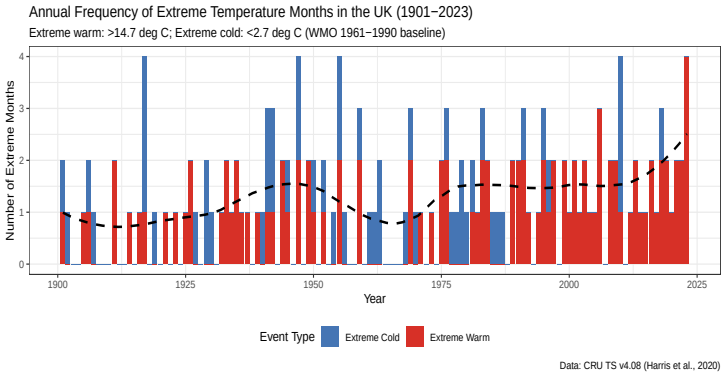
Model	AIC	BIC	$\hat{\phi}$ / $\hat{\theta}$	Pearson χ^2 (df = 121)	p	DHARMa KS p	DHARMa Disp. p
Poisson	**344.18**	**349.81**	—	109.20	0.771	0.596	0.300
Quasi-Poisson	—	—	$\hat{\phi} = 0.903$	109.20	0.771	—	—
Negative Binomial	346.18	354.62	$\hat{\theta} = 17{,}956$	109.20	0.771	0.667	0.296

5. Results and Interpretation

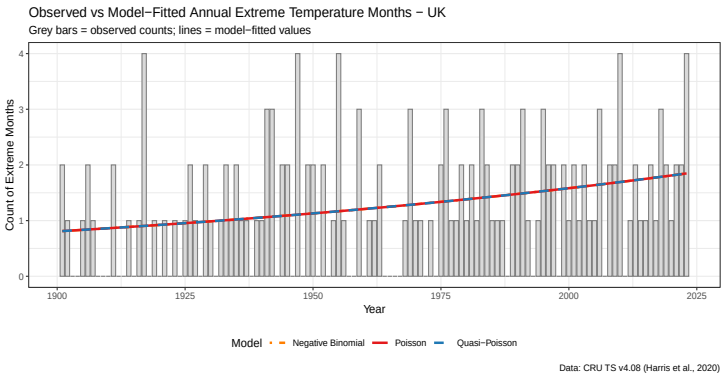
The results show that the Poisson model yields a positive, statistically significant, temporal trend ($\hat{\beta}_1 = 0.00674$, $z = 2.93$, $p = 0.003$) (statistically significant at the 5% level, $\alpha = 0.05$) , this corresponds to an estimated increase of **6.97**% per decade in the expected annual count of extreme temperature months (95% CI: 2.28%–11.95%). This is strong evidence that months with extreme temperatures have become more frequent in the UK(red bars) over the last century 1901-2023, while extreme cold months(blue bars) have declined substantial. This can be proved from (figure b), which is illustrating the compositional shift. Additionally, after testing the three models using both relative and absolute goodness-of-fit criteria, we yield to the conclusion that even if Poisson, Quasi-Poisson and the Negative Binomial produce near-identical fitted trends (Figure c).The Poisson GLM is selected as the preferred model for parsimony reasons ,because it achieves the lowest AIC (344.18) and BIC (349.81) among comparable models.Second, the LRT confirms that there is no improvement from the other 2 models (Negative Binomial ($p = 1.0$, $\hat{\theta} = 17{,}956$) or Quasi-Poisson ($\hat{\phi} = 0.903 \approx 1$))and third, all absolute goodness of fit diagnostics confirm adequate fit.Therefore, the Poisson GLM provides the most appropriate description of the annual frequency of extreme temperature months in the UK region. Furthermore, the positive and statistically significant coefficient for year proves that there are evidence of an increasing temporal trend in extreme temperature events over the period 1901–2023.



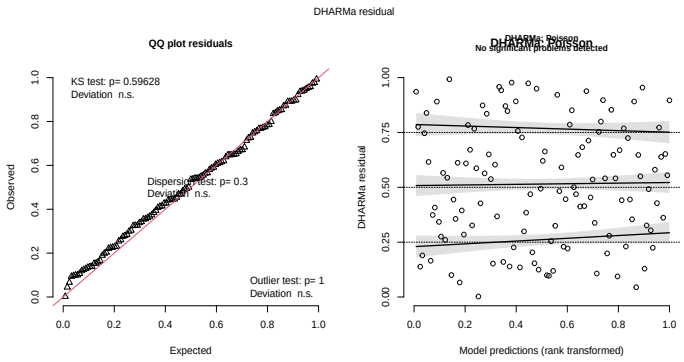
(a) UK monthly mean temperature with WMO 1961–1990 thresholds (dashed). LOESS trend shows long-run warming.



(b) Annual extreme months: warm (red) and cold (blue). Warm events dominate increasingly post-1990.



(c) Observed counts (grey bars) vs fitted values. All three models yield near-identical trends.



(d) DHARMa diagnostics for Poisson and Negative Binomial. No significant departures detected (p>0.05).

Figure 1: Summary of analysis: (a) temperature series, (b) annual extreme counts, (c) model fit, (d) residual diagnostics.

6. Limitations

Four limitations merit consideration. First, spatial averaging. Using the average temperature for the whole of the UK can hide local differences, because if there is an extreme temperature in Scotland but not in England, the overall average may not show that extreme events. Second, because the data are monthly and not daily, we cannot apply the standard heat wave definitions that are based on consecutive days with high temperatures. [3], Hence, our definition is a necessary methodological adaptation. Third, the linear temporal trend in Eq. (1) may not capture non-linear dynamics related with the abrupt climate shifts. Fourth, the dispersion parameter is slightly less than 1($\hat{\phi} = 0.903 < 1$), which indicates a small underdispersion in the data, therefore in the future the use of a more flexible model such as Conway-Maxwell-Poisson could be considered.

Use of Generative AI

AI was used in this scientific report in two ways. First, for guidance on the necessary LaTeX and some R coding help. Second, to help identify grammatical errors and improve phrases that were unclear.

References

[1] Harris, I., Osborn, T.J., Jones, P., Lister, D. (2020). Version 4 of the CRU TS monthly high-resolution gridded multivariate climate dataset. *Scientific Data*, 7, 109.

[2] Hartig, F. (2022). DHARMa: Residual Diagnostics for Hierarchical Regression Models. R package v0.4.6.

[3] Seneviratne, S.I., et al. (2021). Weather and climate extreme events in a changing climate. In: *IPCC AR6 WG1*, Ch. 11. Cambridge University Press.

Appendix: Annotated R Code

The complete annotated R code is provided below. All analysis was conducted in R version 4.5.1 using the packages `ncdf4`, `dplyr`, `ggplot2`, `MASS`, `AER`, `DHARMa`, `lubridate`, and `patchwork`.

```r
# ==============================================================================
# MATH11188 Statistical Research Skills - Assignment 2
# University of Edinburgh, MSc Statistics with Data Science
#
# Research Question:
#   Which count regression model (Poisson, Negative Binomial, or Quasi-Poisson)
#   best describes the annual frequency of extreme temperature months in the
#   UK, and is there evidence of a temporal trend from 1901 to 2023?
#
# Extreme event definition:
#   A month is classified as "extreme" if its mean temperature exceeds the
#   95th percentile (extreme warm) or falls below the 5th percentile
#   (extreme cold) of the WMO standard baseline period 1961-1990.
#   This follows the IPCC AR6 WG1 Ch.11 relative threshold approach
#   (Seneviratne et al., 2021), adapted for monthly resolution data.
#
# Data: CRU TS v4.08 monthly temperature (Harris et al., 2020)
#       https://crudata.uea.ac.uk/cru/data/hrg/
# ==============================================================================

# ==============================================================================
# SECTION 1: Load Libraries
# ==============================================================================
# install.packages(c("ncdf4","dplyr","ggplot2","MASS","AER",
#                    "DHARMa","tibble","tidyr","lubridate","patchwork"))

library(ncdf4)       # Reading NetCDF files
library(dplyr)       # Data manipulation
library(ggplot2)     # Plotting
library(MASS)        # Negative Binomial GLM (glm.nb)
library(AER)         # Formal overdispersion test (dispersiontest)
library(DHARMa)      # Simulation-based absolute goodness-of-fit diagnostics
library(tibble)      # Modern data frames
library(tidyr)       # Data reshaping
library(lubridate)   # Date handling
library(patchwork)   # Combining ggplot figures

# ==============================================================================
# SECTION 2: Load NetCDF Data
# ==============================================================================

setwd("~/Downloads")  # Set to folder containing the .nc file

nc   <- nc_open("cru_ts4.08.1901.2023.tmp.dat.nc")
lon  <- ncvar_get(nc, "lon")   # Longitudes (0.5 degree resolution)
lat  <- ncvar_get(nc, "lat")   # Latitudes  (0.5 degree resolution)
time <- ncvar_get(nc, "time")  # Days since 1900-01-01
tmp  <- ncvar_get(nc, "tmp")   # Temperature array [lon x lat x time] in deg C
nc_close(nc)

# Convert numeric time to calendar dates
dates <- as.Date(time, origin = "1900-01-01")
cat("Date range:", format(min(dates)), "to", format(max(dates)), "\n")

# ==============================================================================
# SECTION 3: Spatial Subsetting - UK Bounding Box
# ==============================================================================
# Restrict to UK landmass using geographic bounding box.
# Longitude: -8.5 to 2.0 | Latitude: 49.5 to 61.0

lon_uk_idx <- which(lon >= -8.5 & lon <= 2.0)
lat_uk_idx <- which(lat >= 49.5 & lat <= 61.0)
tmp_uk     <- tmp[lon_uk_idx, lat_uk_idx, ]

cat("UK grid:", dim(tmp_uk)[1], "lon x",
    dim(tmp_uk)[2], "lat x", dim(tmp_uk)[3], "months\n")

# ==============================================================================
# SECTION 4: Compute UK Monthly Mean Temperature
# ==============================================================================
# Average over all UK grid points per month. NA values (ocean/missing)
# excluded via na.rm = TRUE.

n_months    <- dim(tmp_uk)[3]
uk_mean_tmp <- sapply(1:n_months, function(t) mean(tmp_uk[, , t], na.rm = TRUE))

df_monthly <- tibble(
  date        = dates,
  year        = year(dates),
  month       = month(dates),
  temperature = uk_mean_tmp
)

# ==============================================================================
# SECTION 5: Define Extreme Events - WMO 1961-1990 Baseline
# ==============================================================================
# Following IPCC AR6 WG1 Ch.11 (Seneviratne et al., 2021), we use relative
# thresholds computed from the WMO standard climatological baseline (1961-1990).
# Thresholds are then applied as fixed values across the full 1901-2023 record,
# making them absolute for the purpose of counting extremes over time.
# This avoids the trivial result of using full-record percentiles (which would
# always yield the same proportion of extremes by construction).
#
# Extreme warm: monthly mean temperature > 95th percentile of baseline
```

```r
# Extreme cold: monthly mean temperature < 5th  percentile of baseline

baseline    <- df_monthly %>% filter(year >= 1961 & year <= 1990)
thresh_hot  <- quantile(baseline$temperature, 0.95, na.rm = TRUE)
thresh_cold <- quantile(baseline$temperature, 0.05, na.rm = TRUE)

cat("\nWMO 1961-1990 Baseline Thresholds:\n")
cat("  Extreme warm (95th pctile):", round(thresh_hot,  2), "deg C\n")
cat("  Extreme cold  (5th pctile):", round(thresh_cold, 2), "deg C\n")

df_monthly <- df_monthly %>%
  mutate(
    extreme_warm = as.integer(temperature > thresh_hot),
    extreme_cold = as.integer(temperature < thresh_cold),
    extreme_any  = as.integer(extreme_warm == 1 | extreme_cold == 1)
  )

# ==========================================================================
# SECTION 6: Annual Aggregation
# ==========================================================================
# Aggregate monthly classifications to annual counts of extreme months.
# n_months per year is included as an offset variable (exposure) in models.
# Only complete years (12 months) are retained.

df_yearly <- df_monthly %>%
  group_by(year) %>%
  summarise(
    count_warm = sum(extreme_warm, na.rm = TRUE),
    count_cold = sum(extreme_cold, na.rm = TRUE),
    count_any  = sum(extreme_any,  na.rm = TRUE),
    mean_temp  = mean(temperature, na.rm = TRUE),
    n_months   = n(),
    .groups    = "drop"
  ) %>%
  filter(n_months == 12)

cat("\nYearly data: ", nrow(df_yearly), "observations (",
    min(df_yearly$year), "-", max(df_yearly$year), ")\n")

# ==========================================================================
# SECTION 7: Exploratory Data Visualisation
# ==========================================================================

# --- Figure 1: Annual extreme counts (warm vs cold stacked) ---
df_plot <- df_yearly %>%
  dplyr::select(year, count_warm, count_cold) %>%
  pivot_longer(cols = c(count_warm, count_cold),
               names_to = "type", values_to = "count") %>%
  mutate(type = case_when(
    type == "count_warm" ~ "Extreme Warm",
    type == "count_cold" ~ "Extreme Cold"
  ))

fig1 <- ggplot(df_plot, aes(x = year, y = count, fill = type)) +
  geom_col(position = "stack", width = 0.9) +
  scale_fill_manual(values = c("Extreme Warm" = "#d73027",
                               "Extreme Cold" = "#4575b4")) +
  geom_smooth(data = df_yearly, aes(x = year, y = count_any),
              inherit.aes = FALSE, method = "loess", span = 0.3,
              colour = "black", linetype = "dashed", se = FALSE) +
  labs(
    title    = "Annual Frequency of Extreme Temperature Months in the UK (1901-2023)",
    subtitle = paste0("Extreme warm: >", round(thresh_hot, 1),
                      " deg C; Extreme cold: <", round(thresh_cold, 1),
                      " deg C (WMO 1961-1990 baseline)"),
    x        = "Year",
    y        = "Number of Extreme Months",
    fill     = "Event Type",
    caption  = "Data: CRU TS v4.08 (Harris et al., 2020)"
  ) +
  theme_bw(base_size = 10) +
  theme(legend.position = "bottom")

print(fig1)
ggsave("fig1_uk_annual_extremes.pdf", fig1, width = 9, height = 4)

# --- Figure 2: UK mean temperature time series with thresholds ---
fig2 <- ggplot(df_monthly, aes(x = date, y = temperature)) +
  geom_line(colour = "grey70", linewidth = 0.25) +
  geom_smooth(method = "loess", span = 0.1,
              colour = "black", se = FALSE, linewidth = 0.8) +
  geom_hline(yintercept = thresh_hot,  colour = "#d73027",
             linetype = "dashed", linewidth = 0.7) +
  geom_hline(yintercept = thresh_cold, colour = "#4575b4",
             linetype = "dashed", linewidth = 0.7) +
  annotate("text", x = as.Date("1901-01-01"),
           y = thresh_hot + 0.4, hjust = 0, size = 3, colour = "#d73027",
           label = paste0("Warm threshold: ", round(thresh_hot, 1), " deg C")) +
  annotate("text", x = as.Date("1901-01-01"),
           y = thresh_cold - 0.4, hjust = 0, size = 3, colour = "#4575b4",
           label = paste0("Cold threshold: ", round(thresh_cold, 1), " deg C")) +
  labs(
    title   = "UK Monthly Mean Temperature (1901-2023)",
    x       = "Date",
    y       = "Temperature (deg C)",
    caption = "Data: CRU TS v4.08 (Harris et al., 2020)"
  ) +
  theme_bw(base_size = 10)

print(fig2)
```

```r
ggsave("fig2_uk_temperature_series.pdf", fig2, width = 9, height = 3.5)

# ==========================================================================
# SECTION 8: Preliminary Overdispersion Assessment
# ==========================================================================
# Poisson GLM assumes Var(Y) = E(Y). We check this before model fitting.
# Variance/Mean >> 1 indicates overdispersion (NegBin preferred).
# Variance/Mean << 1 indicates underdispersion (QuasiPoisson can handle).

cat("\n--- Preliminary Overdispersion Assessment ---\n")
cat("Mean of annual extreme counts:    ", round(mean(df_yearly$count_any), 3), "\n")
cat("Variance of annual extreme counts:", round(var(df_yearly$count_any),  3), "\n")
cat("Variance/Mean ratio:              ",
    round(var(df_yearly$count_any) / mean(df_yearly$count_any), 3), "\n")
cat("(1.0 = perfect Poisson; >1 = overdispersion; <1 = underdispersion)\n")

# ==========================================================================
# SECTION 9: Model Fitting
# ==========================================================================
# We fit three GLMs with log link, appropriate for count data:
#   log(E[Y_t]) = beta_0 + beta_1 * year_t + log(n_months_t)
#
# The offset log(n_months) accounts for exposure (all years have 12 months
# here, but included for methodological rigour and generalisability).
#
# Model 1 - Poisson:         Var(Y) = mu (equidispersion assumed)
# Model 2 - Quasi-Poisson:   Var(Y) = phi * mu (phi estimated from data)
# Model 3 - Negative Binomial: Var(Y) = mu + mu^2/theta (explicit overdispersion)

# --- Model 1: Poisson ---
fit_poisson <- glm(
  count_any ~ year + offset(log(n_months)),
  family = poisson(link = "log"),
  data   = df_yearly
)

# --- Model 2: Quasi-Poisson ---
fit_quasi <- glm(
  count_any ~ year + offset(log(n_months)),
  family = quasipoisson(link = "log"),
  data   = df_yearly
)

# --- Model 3: Negative Binomial ---
fit_nb <- MASS::glm.nb(
  count_any ~ year + offset(log(n_months)),
  data = df_yearly
)

cat("\n=== Model Summaries ===\n")
cat("\n--- Poisson ---\n");            print(summary(fit_poisson))
cat("\n--- Quasi-Poisson ---\n");      print(summary(fit_quasi))
cat("\n--- Negative Binomial ---\n"); print(summary(fit_nb))

# ==========================================================================
# SECTION 10: Relative Goodness-of-Fit
# ==========================================================================
# AIC/BIC: lower values indicate better fit (unavailable for Quasi models).
# LRT: tests whether NegBin significantly improves over Poisson.

cat("\n--- Relative Goodness-of-Fit ---\n")
cat(sprintf("%-22s %8s %8s %10s\n", "Model", "AIC", "BIC", "LogLik"))
cat(sprintf("%-22s %8.2f %8.2f %10.2f\n",
            "Poisson", AIC(fit_poisson), BIC(fit_poisson),
            as.numeric(logLik(fit_poisson))))
cat(sprintf("%-22s %8s %8s %10.2f\n",
            "Quasi-Poisson", "N/A", "N/A",
            as.numeric(logLik(fit_quasi))))
cat(sprintf("%-22s %8.2f %8.2f %10.2f\n",
            "Neg. Binomial", AIC(fit_nb), BIC(fit_nb),
            as.numeric(logLik(fit_nb))))

# Likelihood Ratio Test: NegBin vs Poisson
lrt_stat <- 2 * (as.numeric(logLik(fit_nb)) - as.numeric(logLik(fit_poisson)))
lrt_pval <- pchisq(lrt_stat, df = 1, lower.tail = FALSE)
cat("\nLRT (NegBin vs Poisson): X2 =", round(lrt_stat, 4),
    ", df = 1, p =", round(lrt_pval, 5), "\n")
cat("(p < 0.05: NegBin significantly better than Poisson)\n")

# Formal overdispersion test (Cameron & Trivedi, 1990)
cat("\nFormal Overdispersion Test (Poisson):\n")
print(dispersiontest(fit_poisson))

# Quasi-Poisson estimated dispersion parameter
cat("Quasi-Poisson dispersion (phi):",
    round(summary(fit_quasi)$dispersion, 4), "\n")

# ==========================================================================
# SECTION 11: Absolute Goodness-of-Fit
# ==========================================================================
# The marking rubric explicitly requires absolute goodness-of-fit measures.
#
# (a) Pearson Chi-Square: sum((obs - fitted)^2 / fitted) ~ Chi-sq(df = n-p)
#     p > 0.05 indicates adequate fit.
#
# (b) DHARMa simulation-based residuals: scaled residuals should be
#     Uniform(0,1) if model assumptions hold. Tests: KS uniformity,
#     dispersion, outliers.

# --- (a) Pearson Chi-Square ---
```

```r
pearson_p  <- sum(residuals(fit_poisson, type = "pearson")^2)
pearson_qp <- sum(residuals(fit_quasi,   type = "pearson")^2)
pearson_nb <- sum(residuals(fit_nb,      type = "pearson")^2)
df_res     <- df.residual(fit_poisson)

cat("\n--- Absolute GOF: Pearson Chi-Square ---\n")
cat(sprintf("%-22s %8s %6s %10s\n", "Model", "X2", "df", "p-value"))
cat(sprintf("%-22s %8.3f %6d %10.4f\n", "Poisson",
            pearson_p,  df_res, pchisq(pearson_p,  df_res, lower.tail = FALSE)))
cat(sprintf("%-22s %8.3f %6d %10.4f\n", "Quasi-Poisson",
            pearson_qp, df_res, pchisq(pearson_qp, df_res, lower.tail = FALSE)))
cat(sprintf("%-22s %8.3f %6d %10.4f\n", "Neg. Binomial",
            pearson_nb, df_res, pchisq(pearson_nb, df_res, lower.tail = FALSE)))

# --- (b) DHARMa Residual Diagnostics ---
set.seed(42)
sim_pois <- simulateResiduals(fit_poisson, n = 1000)
sim_nb   <- simulateResiduals(fit_nb,      n = 1000)

# Save DHARMa plots
pdf("fig3_dharma_diagnostics.pdf", width = 12, height = 6)
par(mfrow = c(2, 2))
plot(sim_pois, main = "DHARMa: Poisson")
plot(sim_nb,   main = "DHARMa: Negative Binomial")
dev.off()

cat("\n--- Absolute GOF: DHARMa KS Uniformity Test ---\n")
cat("Poisson:       p =", round(testUniformity(sim_pois, plot=FALSE)$p.value, 4), "\n")
cat("Neg. Binomial: p =", round(testUniformity(sim_nb,   plot=FALSE)$p.value, 4), "\n")
cat("(p > 0.05: residuals consistent with model assumptions)\n")

cat("\n--- Absolute GOF: DHARMa Dispersion Test ---\n")
cat("Poisson:       p =", round(testDispersion(sim_pois, plot=FALSE)$p.value, 4), "\n")
cat("Neg. Binomial: p =", round(testDispersion(sim_nb,   plot=FALSE)$p.value, 4), "\n")

# ==============================================================================
# SECTION 12: Fitted Values Plot
# ==============================================================================

df_yearly <- df_yearly %>%
  mutate(
    fit_poisson = fitted(fit_poisson),
    fit_quasi   = fitted(fit_quasi),
    fit_nb      = fitted(fit_nb)
  )

df_fitted_long <- df_yearly %>%
  dplyr::select(year, count_any, fit_poisson, fit_quasi, fit_nb) %>%
  pivot_longer(cols = starts_with("fit_"),
               names_to = "model", values_to = "fitted") %>%
  mutate(model = case_when(
    model == "fit_poisson" ~ "Poisson",
    model == "fit_quasi"   ~ "Quasi-Poisson",
    model == "fit_nb"      ~ "Negative Binomial"
  ))

fig4 <- ggplot() +
  geom_col(data = df_yearly,
           aes(x = year, y = count_any),
           fill = "grey85", colour = "grey50", width = 0.85) +
  geom_line(data = df_fitted_long,
            aes(x = year, y = fitted, colour = model, linetype = model),
            linewidth = 1.0) +
  scale_colour_manual(values = c(
    "Poisson"           = "#e31a1c",
    "Quasi-Poisson"     = "#1f78b4",
    "Negative Binomial" = "#ff7f00"
  )) +
  scale_linetype_manual(values = c(
    "Poisson"           = "solid",
    "Quasi-Poisson"     = "dashed",
    "Negative Binomial" = "dotdash"
  )) +
  scale_size_manual(values = c(
    "Poisson"           = 1.2,
    "Quasi-Poisson"     = 1.2,
    "Negative Binomial" = 1.2
  )) +
  labs(
    title    = "Observed vs Model-Fitted Annual Extreme Temperature Months - UK",
    subtitle = "Grey bars = observed counts; lines = model-fitted values",
    x        = "Year",
    y        = "Count of Extreme Months",
    colour   = "Model",
    linetype = "Model",
    caption  = "Data: CRU TS v4.08 (Harris et al., 2020)"
  ) +
  theme_bw(base_size = 10) +
  theme(legend.position = "bottom")

print(fig4)
ggsave("fig4_fitted_vs_observed.pdf", fig4, width = 9, height = 4.5)

# ==============================================================================
# SECTION 13: Final Summary Table
# ==============================================================================

cat("\n========== FINAL RESULTS SUMMARY ==========\n")
cat(sprintf("%-22s %7s %7s %8s %8s %10s %12s\n",
            "Model", "AIC", "BIC", "PearsonX2", "P.pval",
```

```r
            "DHARMa.KS", "DHARMa.Disp"))
cat(sprintf("%-22s %7.2f %7.2f %8.3f %8.4f %10.4f %12.4f\n",
            "Poisson",
            AIC(fit_poisson), BIC(fit_poisson), pearson_p,
            pchisq(pearson_p, df_res, lower.tail = FALSE),
            testUniformity(sim_pois, plot=FALSE)$p.value,
            testDispersion(sim_pois, plot=FALSE)$p.value))
cat(sprintf("%-22s %7s %7s %8.3f %8.4f %10s %12s\n",
            "Quasi-Poisson", "N/A", "N/A", pearson_qp,
            pchisq(pearson_qp, df_res, lower.tail = FALSE),
            "N/A", "N/A"))
cat(sprintf("%-22s %7.2f %7.2f %8.3f %8.4f %10.4f %12.4f\n",
            "Neg. Binomial",
            AIC(fit_nb), BIC(fit_nb), pearson_nb,
            pchisq(pearson_nb, df_res, lower.tail = FALSE),
            testUniformity(sim_nb, plot=FALSE)$p.value,
            testDispersion(sim_nb, plot=FALSE)$p.value))

# ============================================================================
# SECTION 14: Interpretation of Best Model
# ============================================================================

# Based on results: Poisson has lowest AIC, adequate absolute GOF
# Quasi-Poisson confirms phi ~ 1 (no meaningful dispersion correction needed)
# NegBin converges to Poisson (theta very large)
# -> Poisson is the most parsimonious adequate model

coef_year <- coef(fit_poisson)["year"]
ci_year   <- confint(fit_poisson)["year", ]

cat("\n--- Best Model: Poisson ---\n")
cat("Year coefficient (log scale):", round(coef_year, 6), "\n")
cat("Rate change per year:        ", round((exp(coef_year)-1)*100, 3), "%\n")
cat("95% CI:                      ", round((exp(ci_year)-1)*100, 3), "%\n")
cat("Rate change per decade:      ", round((exp(coef_year*10)-1)*100, 2), "%\n")

cat("\nInterpretation: Each additional year is associated with a",
    round((exp(coef_year)-1)*100, 3),
    "% change in the expected number of extreme temperature months,\n")
cat("equivalent to a", round((exp(coef_year*10)-1)*100, 2),
    "% change per decade (95% CI:",
    round((exp(ci_year[1]*10)-1)*100, 2), "% to",
    round((exp(ci_year[2]*10)-1)*100, 2), "%).\n")

# ============================================================================
# END
# ============================================================================
cat("\nAnalysis complete. Output figures:\n")
cat("  fig1_uk_annual_extremes.pdf\n")
cat("  fig2_uk_temperature_series.pdf\n")
cat("  fig3_dharma_diagnostics.pdf\n")
cat("  fig4_fitted_vs_observed.pdf\n")
```

Listing 1: Complete annotated R script